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22P0234

التاريخ: 　　　　　　　　　　　　　　　　　　　　　　　　　　　　　　الموضوع:

18.75 / 20

Q1) • for given histogram, the higher intensities are for the paper "background", while the lower intensities are for the writing. The intensities in the middle is the non-uniform shading and unwanted contrast.

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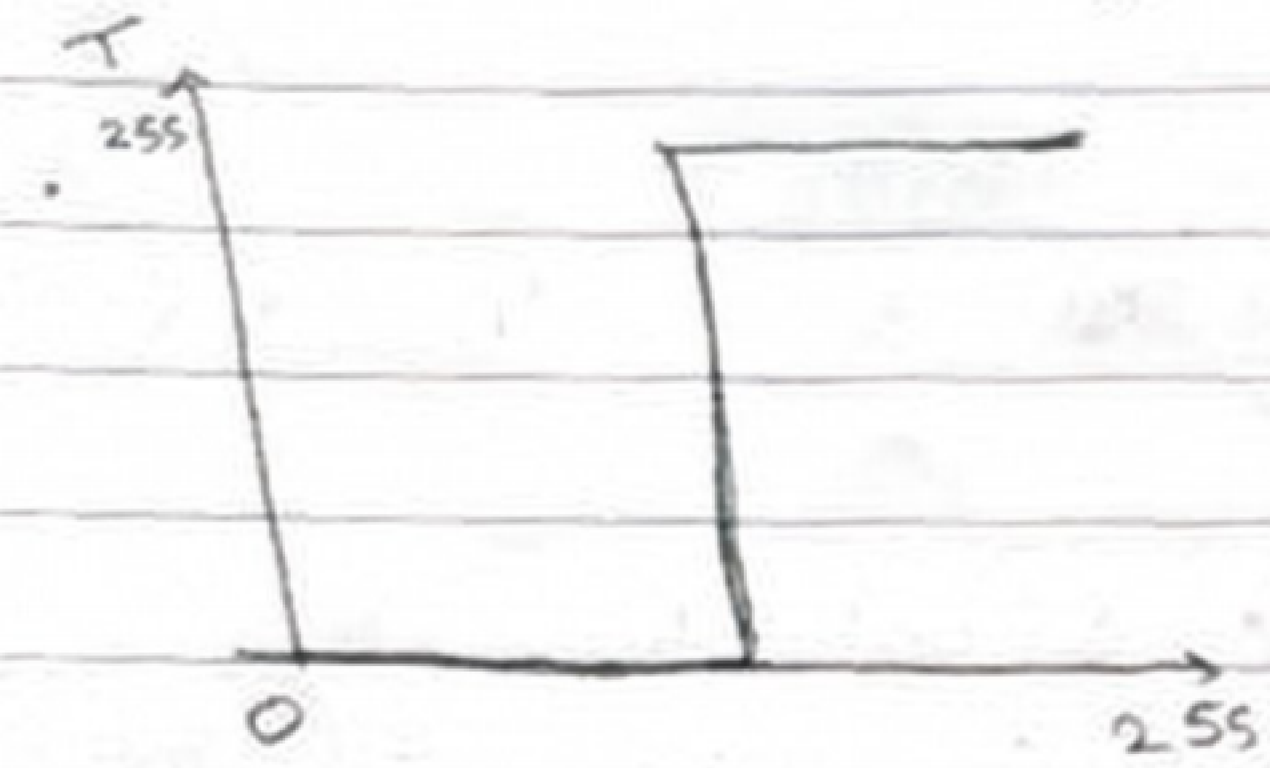
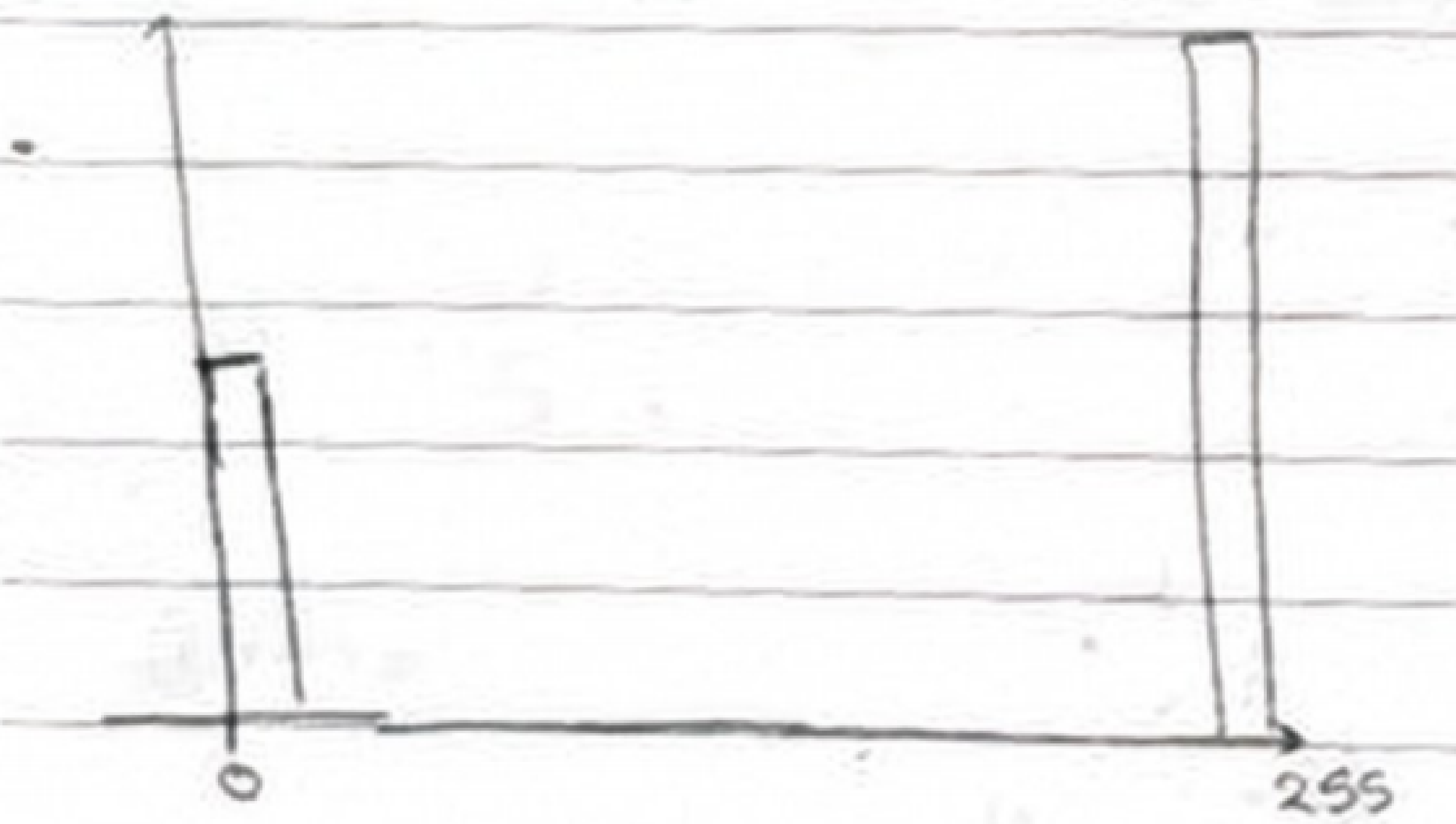
• We will use thresholding to create a binarised image.

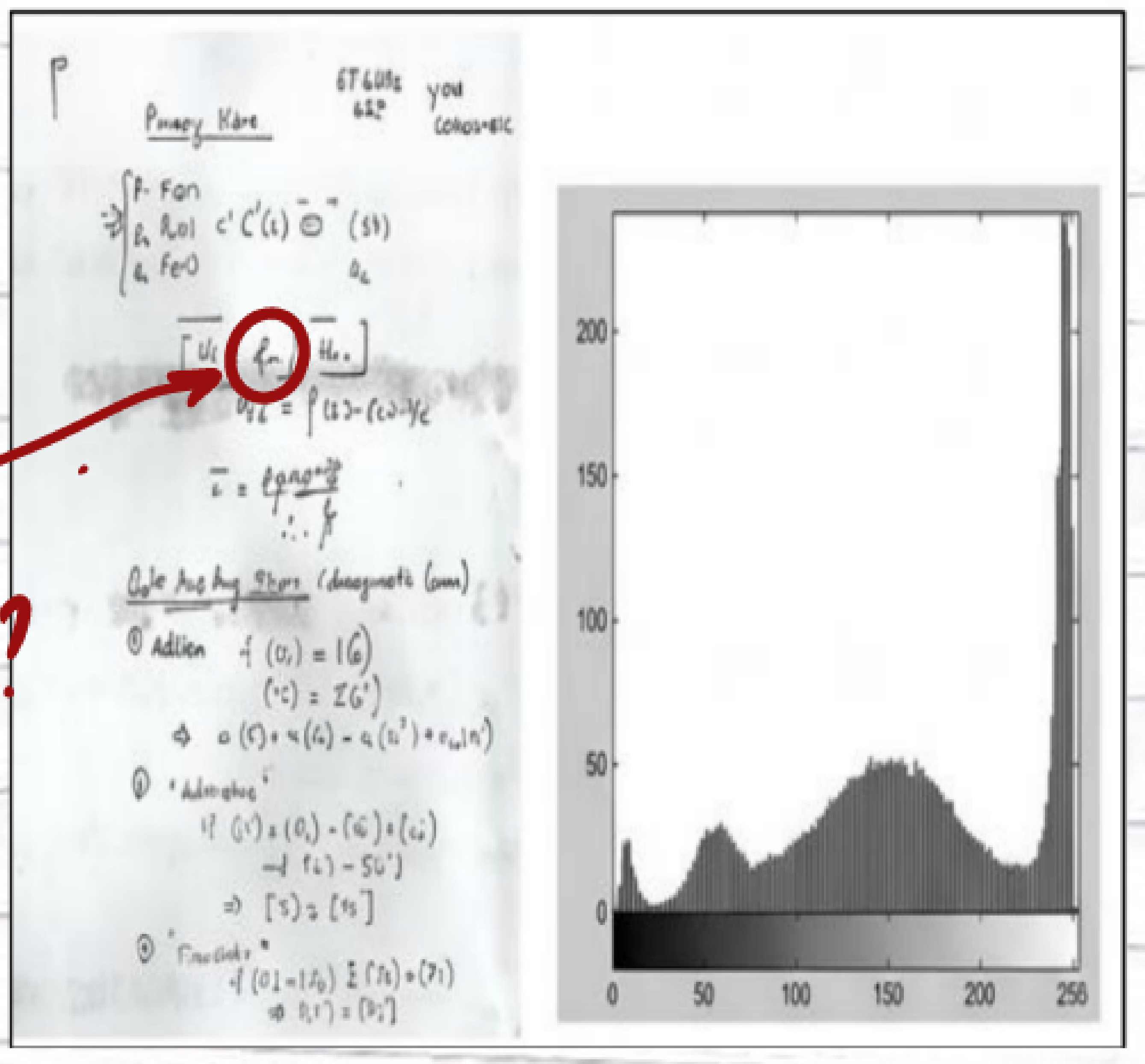


→ intensities below 128 will transform to 0
intensities above 128 will transform to 255



Expected histogram

Shadings have been thresholded along with text
needed to decrease shading first



Transferred to
text or white
background?

الموضوع:

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Q2) A (3,3) , B(7,3), C (7,7) D(3,7)

target A'(6.5,3.4), B'(12.5,3.4), C(12.5,6.6), D'(6.5,6.6)

① first, scale x by 1.5 and y by 4/5

$$S = \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 4/5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

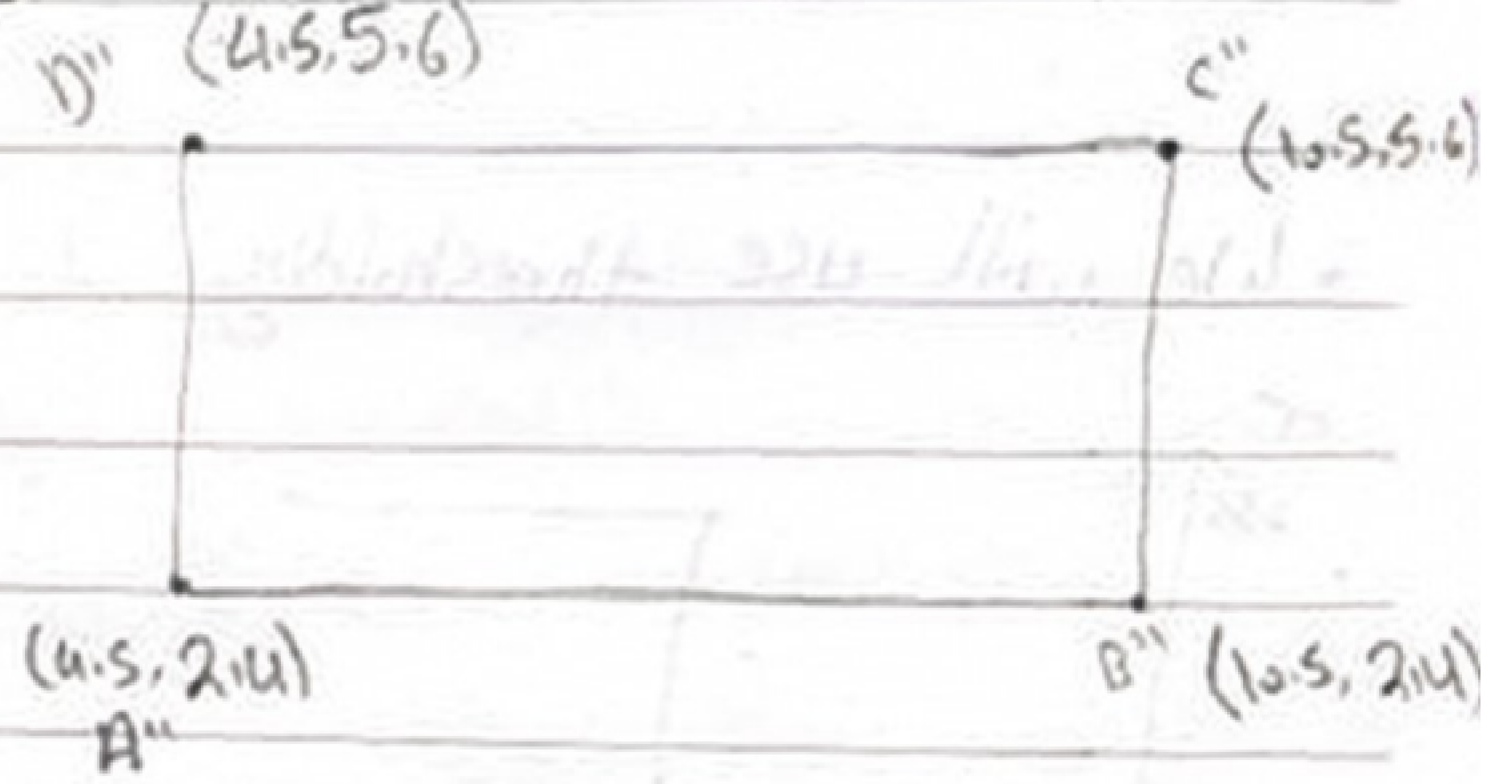
Scaling matrix

$x_A = 4.5$ $y_A = 2.4$

$x_B = 10.5$ $y_B = 2.4$

$x_C = 10.5$ $y_C = 5.6$

$x_D = 4.5$ $y_D = 5.6$



② second, translate 2 units to the right and 1 unit up.

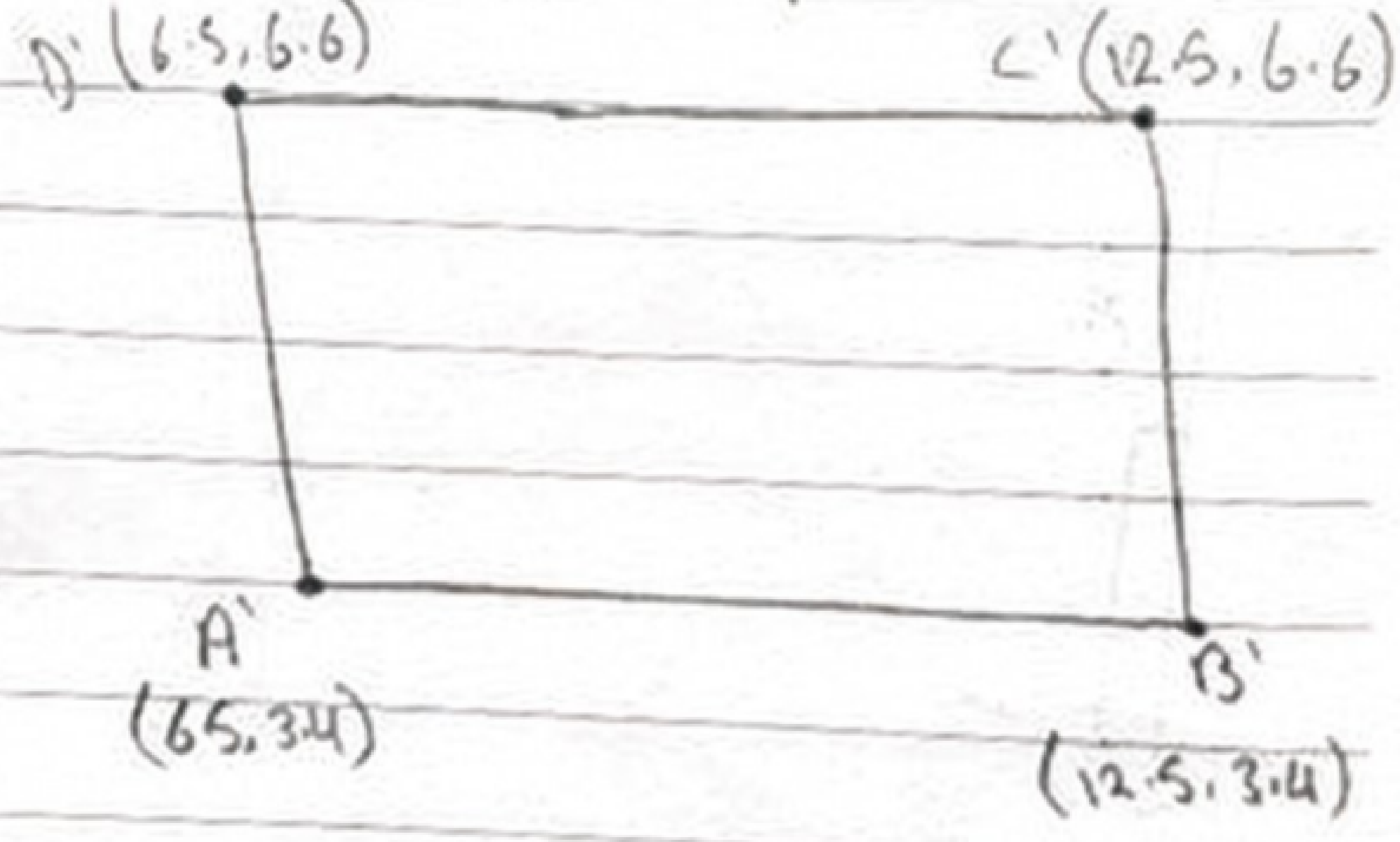
$$T = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$x_A = 6.5$ $y_A = 3.4$

$x_B = 12.5$ $y_B = 3.4$

$x_C = 12.5$ $y_C = 6.6$

$x_D = 6.5$ $y_D = 6.6$



$$T = \begin{bmatrix} 1.5 & 0 & 2 \\ 0 & 4/5 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

✓

3/3

scaling: $\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $c_x = 1.5$ $c_y = 4/5$

translation: $\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 1 & 0 \end{bmatrix}$ $t_x = 2$ $t_y = 1$

Q3) ① GE - TB - H3
② GB - TC - H4
③ GF - TE - H5
④ GD - TA - H2
⑤ GC - TD - H1

unused is CA

8/8

Excellent

Thinking: "Reasons"

① GE is negative transformation, which matches TB
H3 is a mirrored version of original.

② GB is a $\gamma > 1$ trans. which matches TC because it is darkened. H4 is shifted left "darkened".

③ GF is thresholding. TE is a binarized Image.
H5 is either 0 or 255.

④ GD is $\gamma < 1$, brightens up image a little bit "TA"
H2 is shifted slightly to the right

⑤ GC is $\gamma < 1$, brightens a lot. TD is brightened a lot.
H1 is shifted to the right a lot

CA unused as it will not change image.

Q4)

r_k	n_k	$P_{r_k} = \frac{n_k}{32}$	CDF	S_k
0	1	1/32	1/32	0
1	4	4/32	5/32	1
2	8	8/32	13/32	3
3	5	5/32	18/32	4
4	2	2/32	20/32	4
5	2	2/32	22/32	5
6	5	5/32	27/32	6
7	5	5/32	1	7

z_k	n_k	P_{z_k}	CDF	V_k
0	0	0	0	0
1	0	0	0	0
2	4	4/32	4/32	1
3	4	4/32	8/32	2
4	4	4/32	12/32	3
5	4	4/32	16/32	4
6	5	5/32	21/32	5
7	11	11/32	1	7

$S_0 = 7(\text{CDF}_0) = 0.22 \to 0$

$S_1 = 7(\text{CDF}_1) = 1.09 \to 1$

$S_2 = 2.84 \to 3$

$S_3 = 3.9 \to 4$

$S_4 = 4.3 \to 4$

$S_5 = 4.8 \to 5$

$S_6 = 5.9 \to 6$

$S_7 = 7$

$V_0 = 7(\text{CDF}_0) = 0$

$V_1 = 0$

$V_2 = 0.875 \to 1$

$V_3 = 1.75 \to 2$

$V_4 = 2.6 \to 3$

$V_5 = 3.5 \to 4$

$V_6 = 4.6 \to 5$

$V_7 = 7$

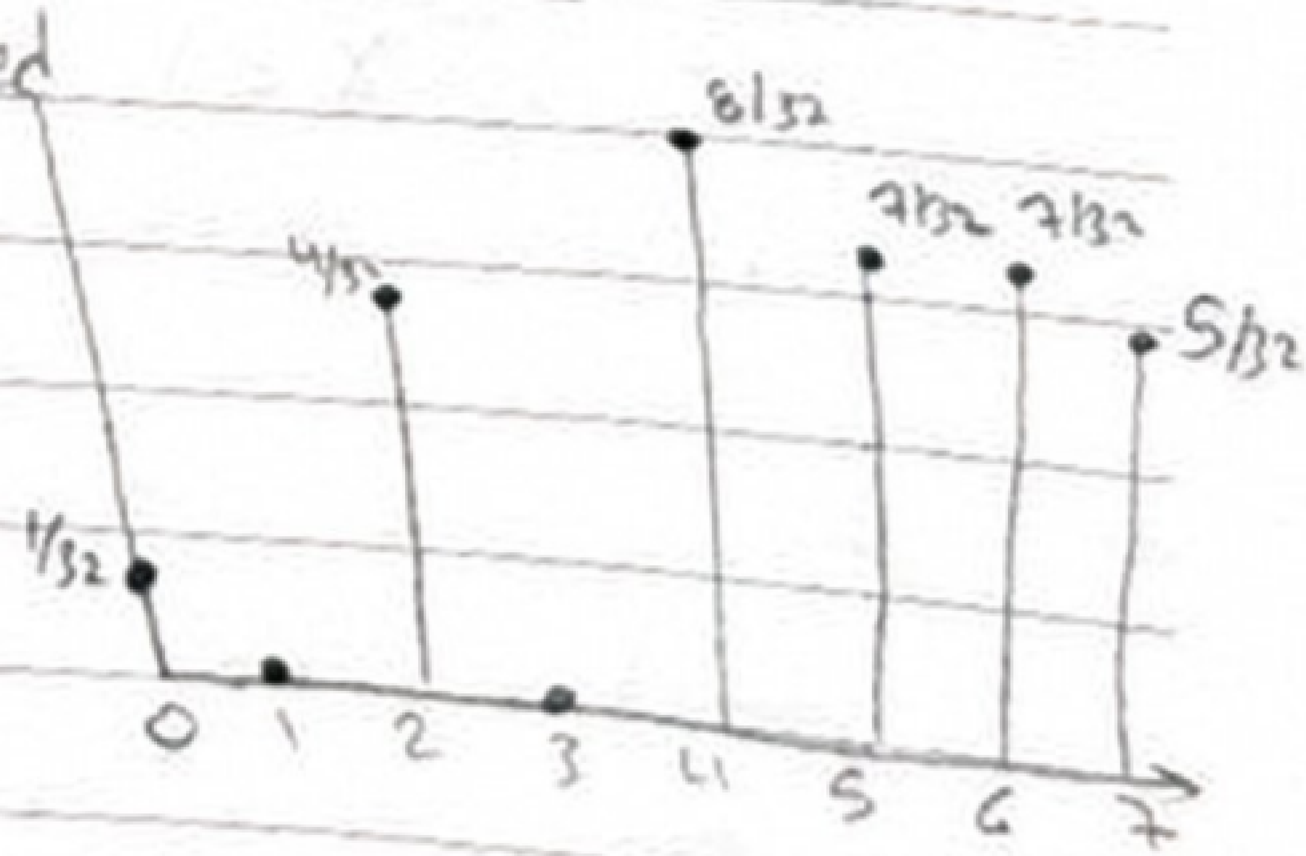
Mapping

r_k	S_k	V_k
0	0 →	0
1	1	0
2	3	1
3	4	2
4	4	3
5	5	4
6	6 →	5
7	7 →	7

Z	
0	1/32
2	4/32
4	8/32
5	5/32
5	7/32
6	7/32
6	5/32
7	5/32

we cannot draw correct image grid for a transformed image as histogram gives no spacial inf.

All we know is how the intensities is now distributed.



−0.5

You have the image itself not the histogram